

Data Challenge Report

Abstract

We propose a face occlusion rate regression model based on DINOv3 ViT-B/16, adapted via the ExPLoRA fine-tuning strategy combining selective block unfreezing and LoRA adapters. A multi-stream regression head fuses global, spatial, texture and saliency features extracted from the backbone into a single occlusion rate prediction. To compensate for the scarcity of heavily occluded samples, we generate a synthetic augmentation dataset using geometrically composited occluders on clean celebrity faces, and train with a 2 phase curriculum designed to balance distributional robustness with target alignment. The full pipeline runs on a consumer GPU and keeps trainable parameters around 20% of the backbone. Results remain in the range of competing approaches, though synthetic data quality appears to be the main limiting factor.

Architecture

Backbone :

The backbone is a pretrained DINOv3 ViT-B/16 (87M parameters). Adaptation follows the ExPLoRA strategy (Khanna et al., Stanford, ICML 2025, arXiv:2406.10973): the last two transformer blocks (10–11) are fully unfrozen for finetuning, while blocks 0–9 receive LoRA adapters (Hu et al., ICLR 2022, arXiv:2106.09685) injected on the q, k, v projections. LayerNorm layers across all blocks are also unfrozen to recalibrate activation distributions for the face domain. This brings the effective number of trainable parameters to roughly 16.1M out of 87M (18%) for the back, making the full training run feasible on a consumer grade RTX 4070 Laptop (8 GB VRAM). Memory is further managed through AMP bfloat16 and gradient checkpointing, reducing the VRAM footprint by approximately 40%. Optimization uses AdamW with four distinct learning rate groups and a OneCycleLR cosine scheduler with a 5-epoch warmup.

Regression Decoder :

The model extracts four complementary representations from the backbone. The final CLS token captures global face semantics. A CrossAttentionHead with a learnable query performs cross-attention over the 196 patch tokens (internally projected to 256 dimensions) to spatially localize occluded regions without mean pooling. Patch tokens from the intermediate block 5 are mean-pooled into a low-frequency texture representation. Finally, CLS-to-patch attention maps from the last block are aggregated by an AttentionMapEncoder, explicitly skipping the 4 DINOv3 register tokens that would otherwise absorb roughly 85% of CLS attention and corrupt the saliency signal, as identified by Darcet et al. (ICLR 2024, arXiv:2309.16588). The four streams are projected to dimension 256, augmented with learned stream position embeddings, and fused by a 2 layer Transformer decoder before a progressive regression head. The loss is identical to the challenge metric and directly penalizes the gender error gap during training.

Data

Data augmentation et sampling :

Online augmentations are carefully selected to avoid label inconsistencies, for example horizontal flip and RandomErasing are explicitly excluded, as both modify the visual content without updating the geometric occlusion GT. Retained transforms include ColorJitter, random grayscale (20%), GaussianBlur, Gaussian noise, JPEG compression artifacts, low resolution simulation and salt-and-pepper noise (selected based on the observation that some images contain noisy patterns which do not seem to have any impact on the ground truth (GT)) . A Weighted Random Sampler combining gender and occlusion axes simultaneously rebalances each batch : gender weights are inversely proportional to class frequency, and occlusion weights are computed over 6 quantile bins. The validation set always uses the original dataset, never augmented data.

Generation d'un dataset additionel :

The synthetic pipeline is inspired by Voo et al. (CVPRW 2022, arXiv:2205.06218), who demonstrate that CelebAMask-HQ combined with COCO objects and DTD textures enables effective synthetic occlusion generation robust to unseen occlusion types. The pipeline runs in three steps. First, the 30,000 CelebAMask-HQ images are filtered by structural criteria (excluding images with non-empty eye_g or hat masks) and then by the pretrained model itself, retaining only faces predicted below 2% occlusion. Second, occluders from COCO (with binary masks) and DTD (texture patches) are composited onto clean faces using fused facial segmentation masks, with the GT computed geometrically as the intersection of occluder and face pixels divided by total face pixels. Third, images are generated across 8 occlusion bins from 20% to 100%, balanced by gender, and integrated into the training set with controlled per-bin sampling to avoid drowning the original signal.

Training follows a 2 phase curriculum. Phase 1 (epochs ~ 30) trains on the combined dataset (original + synthetic), exposing the model to a wider and more uniform occlusion distribution to build robustness against unknown test distribution shifts. The sampler ensures balanced batches throughout. Phase 2 (~ 20 epoch) automatically switches to the original dataset only, removing all synthetic data and re-centering the model on the true target distribution. This mirrors a pretraining then fine-tuning logic, applied at the data level rather than the parameter level.

Conclusion

Results remain in the same order of magnitude as other competing groups but are relatively disappointing given the complexity of the approach. The most likely explanation lies in the synthetic data strategy: despite bin and gender balancing, the ~18,800 generated images appear to have introduced excessive distributional noise during Phase 1. COCO objects and DTD textures composited geometrically remain distant from the naturalistic occlusions present in the target dataset, and the model likely over-adapted to artificial patterns that do not transfer well. A shorter Phase 1 or a lower synthetic-to-original ratio would probably have better preserved the target distribution signal. On the model side, while the architecture may appear heavyweight at first glance, the use of LoRA and limited block unfreezing keeps the effectively trainable parameter count at roughly 18.3M (backbone + decoder) out of 89M, enabling a full training run on a consumer RTX 4070 Laptop in under a day and an half

Here is the link to the github : https://github.com/Nat798/Rogue_One_A_challenged_data_story.git

Source

Scientific litterature :

Delving into High-Quality Synthetic Face Occlusion Segmentation Datasets

Voo et al., CVPRW 2022

<https://arxiv.org/abs/2205.06218>

FaceOcc — A Diverse, High-quality Face Occlusion Dataset for Human Face Extraction

Yin and Chen, TAIMA 2022

<https://arxiv.org/abs/2201.08425>

LoRA — Low-Rank Adaptation of Large Language Models

Hu et al., ICLR 2022

<https://arxiv.org/abs/2106.09685>

ExPLoRA — Parameter-Efficient Extended Pre-Training to Adapt Vision Transformers under Domain Shifts

Khanna et al., Stanford University, ICML 2025

<https://arxiv.org/abs/2406.10973>

Additional dataset :

<https://partage.imt.fr/index.php/s/ntYk27ZFCbeKGqW>

<https://drive.google.com/drive/folders/15nZETWIGMdcKY6aHbchRsWkUI42KTNs5>

<https://www.robots.ox.ac.uk/~vgg/data/dtd/>

<https://www.kaggle.com/datasets/ipythonx/celebamaskhq>